higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION: BOILERMAKING
NQF LEVEL 4

(6031014)

12 March 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

1.1 Make a neat drawing of a pressure vessel and label the following components:

1.1.1 Dished end

1.1.2 Nozzle

1.1.3 Saddle

1.1.4 Access way

1.1.5 Shell

(5 × 1) (5)

1.2 The table below contains the types of supplementary welding symbols.

Complete the table by filling in the missing parts. Write only the answer next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

1.2.1	Site weld	Surface contour			Finish			
		Flush	Convex	Concave	Machine	Grind	Chisel	**1.2.5**
		**1.2.2**	**1.2.3**		**1.2.4**	G	C	F

(5)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

1.3.1 A fillet weld is a type of weld that joins two parts at right angles.

1.3.2 During the hot-rolled process a billet passes through a pair of hot rollers.

1.3.3 The area of a circle is calculated by using the formula π × D.

1.3.4 The mean diameter can be calculated without the thickness of the material.

1.3.5 A datum line is the centre line of a symmetrical component.

1.3.6 A bevel gauge is used to bevel workpieces for welding.

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1.3.7 You can use a pressure vessel without a gauge if it is newly constructed and already pressure tested.

1.3.8 The symbol to indicate level on a drawing is a triangle.

1.3.9 Another name for a column is a stanchion.

1.3.10 Pressure testing equipment must be calibrated before use.

(10 × 1) (10)
[20]

QUESTION 2

2.1 Choose an item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–E) next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1.1	BM	A	diagonal
2.1.2	csk	B	detail
2.1.3	diag	C	dimension
2.1.4	dtl	D	back mark
2.1.5	dim	E	countersunk

(5 × 1) (5)

2.2 Name the force that a column is designed to withstand. (1)

2.3 Study the picture below and answer the questions.

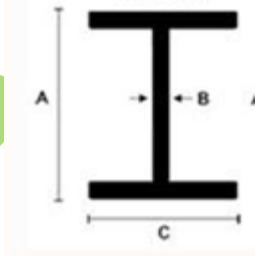


FIGURE 1

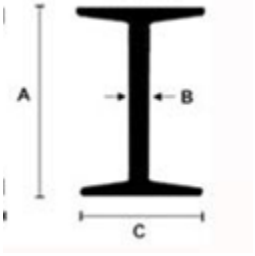


FIGURE 2

2.3.1 Name FIGURE 1 shown above.

2.3.2 Name FIGURE 2 shown above.

2.3.3 Name the plate that is fitted at the top of these columns.

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2.3.4 Name the plate that is fitted at the bottom of these columns. (4 × 1) (4)

2.4 Give THREE reasons why bracings are important in structural work. (3)

2.5 State FOUR kinds of safety that you need to know when working on site. (4)

2.6 Name TWO types of shackles used for rigging on site. (2)

2.7 Which protective clothing must be worn to protect one's hands when lifting, stacking or handling materials? (1)
[20]

QUESTION 3

3.1 Name FIVE types of drawings for pipework fabrication and installation. (5)

3.2 Name FIVE fittings and components used for pipework systems. (5)

3.3 Calculate the travel if the offset is 300 mm and a 30° fitting is used. (5)

3.4 Give FIVE kinds of information obtained from a job card. (5)
[20]

QUESTION 4

4.1 Companies are presently diverting to CNC machines for efficient production.

4.1.1 Name THREE industries which use CNC machines.

4.1.2 Give ONE example of a machine used by each of the industries named in QUESTION 4.1.1. (2 × 3) (6)

4.2 Give SEVEN reasons why a conventional drilling machine operating method is less preferred. (7)

4.3 Give FIVE reasons why a CNC machine operating method is preferred. (5)

4.4 Explain what is meant by the following tool path movements:

4.4.1 Absolute movement

4.4.2 Incremental movement (2 × 2) (4)

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4.5 Explain the meaning of the following codes:

4.5.1 T-codes

4.5.2 S-codes

4.5.3 F-codes

(3 × 2) (6)

4.6 Explain why it is important to simulate the program. (3)

4.7 List FIVE items that you need to consider when you select machines. (5)

4.8 List the THREE factors that you must consider when deciding how much material can be removed during machining. (3)

4.9 Give ONE reason why workpieces are removed from the machine after the first run. (1)

[40]

TOTAL: 100

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